

# Data modalities

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May 2026

An uncomfortable mind search for meaning , happiness , freedom and philosophy of life . Society wants you to be in this state which runs its economy. Thats why you have psychologist, therapist(these people know how to sell their time without a product, actually their product is their time !!!), philosophers(These people know how to keep you in delusion and worldly bias just with words),gurus(These people are the greatest CEOs on earth without MBA, they hire people for work without salary in the name of service and they know how to sell small product for huge margin of profit without ads) and all kind of self development life coaches(dont ever get near to them, they are highly seductive and delusional) , different ways of escape mechanisms like cars , bikes , home , clothes , food , vacation and end less transitory pleasures(these are living proof of human despiration and unhappiness). You run their business through your uncomfortable mind. Its not a problem of society , its doing perfectly fine with its business structure. Its your problem of not recognizing that you are living with an uncomfortable mind. Look what happens when you see this clearly.

A comfortable mind just see the world as it is. And it will fully immerse in present moment without being lost in thoughts , even if it lost , but it knows that it is lost in thoughts. For a comfortable mind there is no difference between any kind of extreme states of life like having phd or being illeterate , having gone to Bali for a vacation or just visited an unknown small road side lake, being high in meditative experience or being high in LSD , watching a adult harmone provoking scenes or watching a discourse of a spiritual guru.

All these are mental events for both comfortable and uncomfortable mind. Only difference is one is carried away with labels and thoughts and the other one is just aware of it.

(Nagesh)

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# Types of Data in Machine Learning

Different machine learning problems involve different underlying data structures.

The geometry and topology of data strongly influence:

- representation learning
- model architecture
- inductive biases

A major theme in deep learning is:

Architecture should respect data structure

Common data types include:

1. Grid Data
2. Sequential Data
3. Graph Data
4. Point Cloud / Set Data
5. Manifold Data
6. Tree Data
7. Spatio-Temporal Data
8. Mesh / Surface Data

## 1. Grid Data

Grid data has regular Euclidean structure.

Examples:

- images
- videos
- volumetric medical scans

## Mathematical Representation

2D image:

$$X \in \mathbb{R}^{H \times W \times C}$$

where:

- $H$  = height
- $W$  = width
- $C$  = channels

Example:

RGB image:

$$C = 3$$

## Properties

- regular lattice structure
- local spatial correlations
- translation symmetry

## Models Used

- CNNs
- Vision Transformers (ViTs)
- UNets
- Diffusion models

## Examples

- MNIST
- CIFAR-10
- MRI scans
- satellite imagery

## 2. Sequential Data

Sequential data has temporal or ordered structure.

Examples:

- text
- speech
- DNA sequences
- financial time series

### Mathematical Representation

Sequence:

$$X = \{x_1, x_2, \dots, x_T\}$$

where:

$$x_t \in \mathbb{R}^d$$

Thus:

$$X \in \mathbb{R}^{T \times d}$$

### Properties

- ordering matters
- causal dependencies
- temporal correlations

### Models Used

- RNNs
- LSTMs
- GRUs
- Transformers
- State-space models

## Examples

- natural language
- ECG signals
- protein sequences
- stock market data

## 3. Graph Data

Graph data represents relational structures.

A graph:

$$G = (V, E)$$

where:

- $V$  = nodes
- $E$  = edges

## Mathematical Representation

Adjacency matrix:

$$A \in \mathbb{R}^{N \times N}$$

where:

$$A_{ij} = 1$$

if edge exists.

Node features:

$$X \in \mathbb{R}^{N \times d}$$

## Properties

- irregular connectivity
- non-Euclidean structure
- relational interactions

## Models Used

- Graph Neural Networks (GNNs)
- Graph Convolutional Networks (GCNs)
- Graph Attention Networks (GATs)
- Message Passing Networks

## Message Passing

Typical GNN update:

$$h_i^{(k+1)} = \phi \left( h_i^{(k)}, \sum_{j \in \mathcal{N}(i)} \psi(h_i^{(k)}, h_j^{(k)}) \right)$$

## Examples

- social networks
- molecular graphs
- citation networks
- transportation networks

## 4. Point Cloud / Set Data

Point clouds are unordered collections of points.

Examples:

- LiDAR scans
- 3D objects
- particle simulations

## Mathematical Representation

Point set:

$$X = \{x_1, x_2, \dots, x_N\}$$

where:

$$x_i \in \mathbb{R}^d$$

Example:



3D point cloud:

$$x_i = (x, y, z)$$

### Important Property

Sets are permutation invariant.

Meaning:

$$\{x_1, x_2\} = \{x_2, x_1\}$$

Ordering should not affect output.

### Models Used

- PointNet
- PointNet++
- Set Transformers
- Deep Sets

### Deep Sets Formulation

Permutation invariant mapping:

$$f(X) = \rho \left( \sum_{i=1}^N \phi(x_i) \right)$$

### Examples

- autonomous driving LiDAR
- protein atomic coordinates
- galaxy distributions

## 5. Manifold Data

Many high-dimensional datasets lie near low-dimensional manifolds.

## Mathematical Idea

Suppose:

$$x \in \mathbb{R}^D$$

but intrinsic dimension:

$$d \ll D$$

Then data lies on manifold:

$$\mathcal{M} \subset \mathbb{R}^D$$

with:

$$\dim(\mathcal{M}) = d$$

## Examples

- natural images
- speech signals
- molecular conformations

## Models Used

- Autoencoders
- Variational Autoencoders
- Diffusion models
- Manifold learning algorithms

## Manifold Learning Methods

- PCA
- t-SNE
- UMAP
- Isomap
- Laplacian Eigenmaps

## Example

Human face images may exist in:

$$\mathbb{R}^{100000}$$

pixel space, but intrinsic manifold dimension may be much smaller.

## 6. Tree Data

Tree data contains hierarchical structure.

Examples:

- syntax trees
- phylogenetic trees
- file systems

## Mathematical Representation

A tree:

$$T = (V, E)$$

with no cycles.

Hierarchical parent-child relationships.

## Recursive Representation

Node representation:

$$h_i = f(x_i, \{h_j : j \in \text{children}(i)\})$$

## Models Used

- Recursive Neural Networks
- Tree-LSTMs
- Hierarchical Transformers

## Examples

- sentence parse trees
- evolutionary trees
- XML/HTML documents

## 7. Spatio-Temporal Data

Spatio-temporal data evolves across both space and time.

Examples:

- videos
- climate systems
- traffic flow
- fluid dynamics

### Mathematical Representation

Suppose:

$$X(t, s)$$

where:

- $t$  = time
- $s$  = spatial coordinate

Example:

$$X \in \mathbb{R}^{T \times H \times W \times C}$$

for videos.

### Properties

- spatial correlations
- temporal evolution
- dynamical behavior

### Models Used

- ConvLSTM
- Video Transformers
- Neural PDE models
- Spatio-temporal GNNs

## Examples

- weather prediction
- surveillance video
- ocean currents
- reaction-diffusion systems

## 8. Mesh / Surface Data

Meshes represent geometric surfaces.

Widely used in:

- computer graphics
- finite element methods
- molecular geometry

## Mathematical Representation

Mesh:

$$M = (V, E, F)$$

where:

- $V$  = vertices
- $E$  = edges
- $F$  = faces

Example:  
triangle mesh:

$$F = (v_i, v_j, v_k)$$

## Properties

- irregular geometry
- local neighborhoods
- surface topology

## Models Used

- MeshCNN
- Geometric Deep Learning
- Graph Neural Networks
- Neural Operators

## Examples

- 3D CAD models
- protein surfaces
- finite element meshes
- human body scans

## Relationship Between Data Geometry and Models

Different architectures encode different symmetries.

| Data Type       | Main Symmetry / Structure |
|-----------------|---------------------------|
| Grid            | Translation symmetry      |
| Sequential      | Temporal causality        |
| Graph           | Relational topology       |
| Point Cloud     | Permutation invariance    |
| Manifold        | Low-dimensional geometry  |
| Tree            | Hierarchical structure    |
| Spatio-temporal | Space-time dynamics       |
| Mesh            | Surface topology          |

## Unified View

Modern machine learning architectures are fundamentally attempts to learn:

|                                          |
|------------------------------------------|
| Transformations respecting data geometry |
|------------------------------------------|

CNNs exploit:

Euclidean grids

Transformers exploit:

sequence interactions

GNNs exploit:

graph topology

Neural operators exploit:

function space dynamics

Thus data geometry determines model architecture.

## Key Insight

Machine learning architectures are deeply connected to:

- topology
- geometry
- symmetry
- structure of data

The success of deep learning largely comes from designing models that respect the intrinsic structure of data manifolds.